

# High-Dose B Vitamin Supplementation and Progression of Subclinical Atherosclerosis

## A Randomized Controlled Trial

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**Background and Purpose**—Although plasma total homocysteine (tHcy) levels are associated with cardiovascular disease, it remains unclear whether homocysteine is a cause or a marker of atherosclerotic vascular disease. We determined whether reduction of tHcy levels with B vitamin supplementation reduces subclinical atherosclerosis progression.

**Methods**—In this double-blind clinical trial, 506 participants 40 to 89 years of age with an initial tHcy  $>8.5$   $\mu\text{mol/L}$  without diabetes and cardiovascular disease were randomized to high-dose B vitamin supplementation (5 mg folic acid+0.4 mg vitamin B<sub>12</sub>+50 mg vitamin B<sub>6</sub>) or matching placebo for 3.1 years. Subclinical atherosclerosis progression across 3 vascular beds was assessed using high-resolution B-mode ultrasonography to measure carotid artery intima media thickness (primary outcome) and multidetector spiral CT to measure aortic and coronary artery calcium (secondary outcome).

**Results**—Although the overall carotid artery intima media thickness progression rate was lower with B vitamin supplementation than with placebo, statistically significant between-group differences were not found ( $P=0.31$ ). However, among subjects with baseline tHcy  $\geq 9.1$   $\mu\text{mol/L}$ , those randomized to B vitamin supplementation had a statistically significant lower average rate of carotid artery intima media thickness progression compared with placebo ( $P=0.02$ ); among subjects with a baseline tHcy  $<9.1$   $\mu\text{mol/L}$ , there was no significant treatment effect (probability value for treatment interaction=0.02). B vitamin supplementation had no effect on progression of aortic or coronary artery calcification overall or within subgroups.

**Conclusion**—High-dose B vitamin supplementation significantly reduces progression of early-stage subclinical atherosclerosis (carotid artery intima media thickness) in well-nourished healthy B vitamin “replete” individuals at low risk for cardiovascular disease with a fasting tHcy  $\geq 9.1$   $\mu\text{mol/L}$ . (*Stroke*. 2009;40:730-736.)

**Key Words:** atherosclerosis ■ computed tomography ■ folate ■ homocysteine ■ intima media thickness ■ randomized controlled trials ■ vitamin B<sub>12</sub> ■ vitamin B<sub>6</sub> ■ folic acid

Fasting plasma total homocysteine (tHcy) and abnormal homocysteine metabolism unmasked by methionine loading are independently associated with cardiovascular disease (CVD).<sup>1–5</sup> Observational studies and meta-analyses indicate that tHcy is a strong, independent graded risk factor for CVD with a 40% to 60% increased risk for each 3 to 5  $\mu\text{mol/L}$  increase in tHcy.<sup>1,2</sup> Cardiovascular risk associated with fasting tHcy substantially increases when plasma levels exceed 8 to 9  $\mu\text{mol/L}$ .<sup>6,7</sup> Additionally, dietary deficiencies of folic

acid, vitamin B<sub>12</sub>, and vitamin B<sub>6</sub> are risk factors for CVD.<sup>8,9</sup> It remains unclear whether tHcy is a cause or a marker of atherosclerotic vascular disease.

Recent trials failed to show reduction of recurrent cardiovascular events with homocysteine-lowering therapy of folic acid and vitamin B<sub>12</sub> with or without vitamin B<sub>6</sub>.<sup>10,11</sup> It remains unknown whether B vitamin supplementation reduces subclinical atherosclerosis progression or CVD in individuals without pre-existing CVD. The B-Vitamin Ath-

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therosclerosis Intervention Trial (BVAIT) was designed to determine the impact of reducing fasting tHcy and postmethionine loading tHcy (PML) with B vitamin supplementation on subclinical atherosclerosis progression in a CVD-free population.

## Methods

### Study Population and Design

BVAIT was a randomized, double-blind, placebo-controlled trial conducted from November 6, 2000, to June 1, 2006. Subjects were men and postmenopausal women  $\geq 40$  years old with fasting tHcy  $\geq 8.5$   $\mu\text{mol/L}$  and no clinical signs/symptoms of CVD. Exclusion criteria were fasting triglycerides  $>5.64$  mmol/L (500 mg/dL), diabetes mellitus or fasting serum glucose  $>6.99$  mmol/L (126 mg/dL), systolic blood pressure  $\geq 160$  mm Hg and/or diastolic blood pressure  $\geq 100$  mm Hg, untreated thyroid disease, creatinine clearance  $<70$  mL/min, life-threatening illness with prognosis  $<5$  years, or  $>5$  alcoholic drinks daily. The University of Southern California Institutional Review Board approved the study protocol; all participants provided written informed consent.

Computer-generated random numbers were used to assign participants to daily supplementation with 5 mg folic acid + 0.4 mg vitamin B<sub>12</sub> + 50 mg vitamin B<sub>6</sub> or matching placebo in one of 2 strata defined by baseline carotid artery intima media thickness (CIMT;  $<0.75$  mm,  $\geq 0.75$  mm). Within each stratum, blocked randomization occurred with a block size of 4. Participants, clinical staff, imaging specialists, and data monitors were masked to treatment assignment.

Clinic visits occurred every 3 months and vital signs, clinical events, diet, and nonstudy medication and supplement/nutraceutical use were ascertained. Treatment adherence was assessed at each visit by pill compliance and every 6 months by measuring tHcy and B vitamin levels. Every 6 months, carotid ultrasonography, oral methionine loading, and fasting blood samples were obtained. A chemistry panel and complete blood count were obtained before randomization and annually.

The initial 2.5-year treatment period was extended on average 1 to 2 years by the External Data and Safety Monitoring Board based on evolving results from secondary prevention B vitamin trials. Interim analyses of the primary trial end point were not performed.

The primary trial end point was the rate of change in the right distal CIMT. Carotid ultrasonography occurred at baseline and every 6 months throughout the original 2.5-year trial period and trial extension. Secondary trial end points were changes in calcium in the coronary arteries (CAC) and abdominal aorta; CT scans were obtained at baseline and at the end of the original 2.5-year trial.

Sample size based on CIMT progression required 176 subjects/arm to detect a moderate effect size of 0.30 at 0.05 significance (2-sided) with 0.80 power. A total of 506 subjects were recruited to accommodate anticipated dropouts and initiation of lipid-lowering medications on-trial.

### Assessment of Atherosclerosis Progression

High-resolution B-mode ultrasound images of the right common carotid artery were obtained with a 7.5-MHz linear array transducer attached to a Toshiba SSH 140A ultrasonography system (Toshiba Corp, Tokyo, Japan). Ultrasound imaging and measurement of far wall CIMT were completed as previously described (patents, 2005, 2006).<sup>12–15</sup> The coefficient of variation of repeated baseline CIMT measurements was  $<1\%$ .

A multidetector spiral CT (Philips Mx-8000 4-S-CT scanner, Cleveland, Ohio) was used to image the coronary arteries and abdominal aorta. Heart scanning began at the carina and proceeded through the cardiac apex; a hydroxyapatite calibration phantom pad was placed under each participant's thorax.<sup>16,17</sup> Simultaneous acquisition of 4 slices and fast-rotation time, restricted breathhold to  $<15$  seconds. Electrocardiographic triggering (set at 50% of the expected next R-R interval) in sequential slice mode at 120 kV and 165 mAs was used to acquire contiguous, noninterlaced slices with a table

increment of 20 mm for every series of 4 slices. Scanning of the abdominal aorta began at the tip of the xyphoid process and proceeded through the level of the umbilicus with a single breathhold. Helical scanning mode at 120 kV and 180 mAs with a table speed of 3 cm/s and pitch of 6 was used for scanning the abdominal aorta and included a calibration phantom pad under each participant's abdomen.<sup>16,17</sup> Scans were analyzed without knowledge of treatment assignment using validated calcium scoring software.<sup>18,19</sup> A separate calcium score for the coronary arteries and abdominal aorta was derived.<sup>20</sup>

### Laboratory Measurements

Participants fasted 8 hours before sample collections. Plasma lipids were measured using an enzymatic method under the Centers for Disease Control and Prevention Standardization Program; low-density lipoprotein cholesterol was calculated.<sup>21</sup> Apolipoprotein A-1 and -B were measured by electroimmunoassay.<sup>22,23</sup> tHcy was determined by reverse-phase high-performance liquid chromatography.<sup>24</sup> Plasma folate and vitamin B<sub>12</sub> were determined by radioimmunoassay kit (Bio-Rad Quanta Phase I and II; Bio-Rad Laboratories, Hercules, Calif). Pyridoxal-5'-phosphate, the active cofactor derived from pyridoxine (vitamin B<sub>6</sub>), was determined enzymatically using a tyrosine decarboxylase-based method.<sup>25</sup> The coefficient of variation for tHcy, folate, vitamin B<sub>12</sub>, and vitamin B<sub>6</sub> measurements were 7.8%, 7%, 7%, and 16%, respectively.

The oral methionine loading test used 100 mg L-methionine/kg body weight in 8 ounces of unsweetened orange/apple juice. After a fasting blood draw, subjects drank the methionine within 5 minutes. Exactly 2 hours after ingestion, the second blood sample was drawn.

An independent laboratory tested each lot of B vitamin pills for content uniformity and dissolution before release as well as stability of the pill components every 3 months for the first year and then every 6 months for Years 2 to 3.

### Statistical Analysis

Prerandomization characteristics were compared between treatment groups with 2-sample *t* tests for continuous variables and  $\chi^2$  tests for categorical variables. Percentage pill compliance, tHcy, and B vitamin levels were averaged over the trial period. Average on-trial levels and changes from baseline were compared between groups with 2-sample *t* tests; changes from baseline were tested within treatment groups using paired *t* tests. PML results were summarized as difference in postload minus fasting tHcy.

An intention-to-treat analysis was performed for all participants who had carotid ultrasonography at baseline and at least one follow-up visit. A linear mixed effects model was used to compare treatment groups on average CIMT change rates. CIMT was regressed on follow-up time (in years) with adjustment for the randomization stratification factor (baseline CIMT). The regression coefficient associated with trial follow-up time estimated the average CIMT annual rate of change. A treatment  $\times$  follow-up time interaction term evaluated whether the treatment groups differed in average CIMT progression rates.

In ancillary analyses, mixed-effects models were used to evaluate the association of baseline and on-trial levels of fasting tHcy, PML, and B vitamins (all modeled as continuous variables) with the rate of CIMT progression. Interaction terms with follow-up time evaluated whether these variables were significantly associated with CIMT progression.

Absolute change in CAC and abdominal aorta was calculated for each subject as: final – baseline calcium score. Treatment group differences were tested using the Wilcoxon rank sum test. Among subjects who had no measurable CAC on the baseline CT scan, the proportion of subjects who developed measurable calcium on the end point scan were compared between treatment groups using Fisher exact test.

Treatment group comparisons of adverse events among all randomized subjects used the Fisher exact test. Adverse events included deaths, cardiovascular events, cerebrovascular events, arterial revascularization procedures, and cancers. The occurrence of white blood

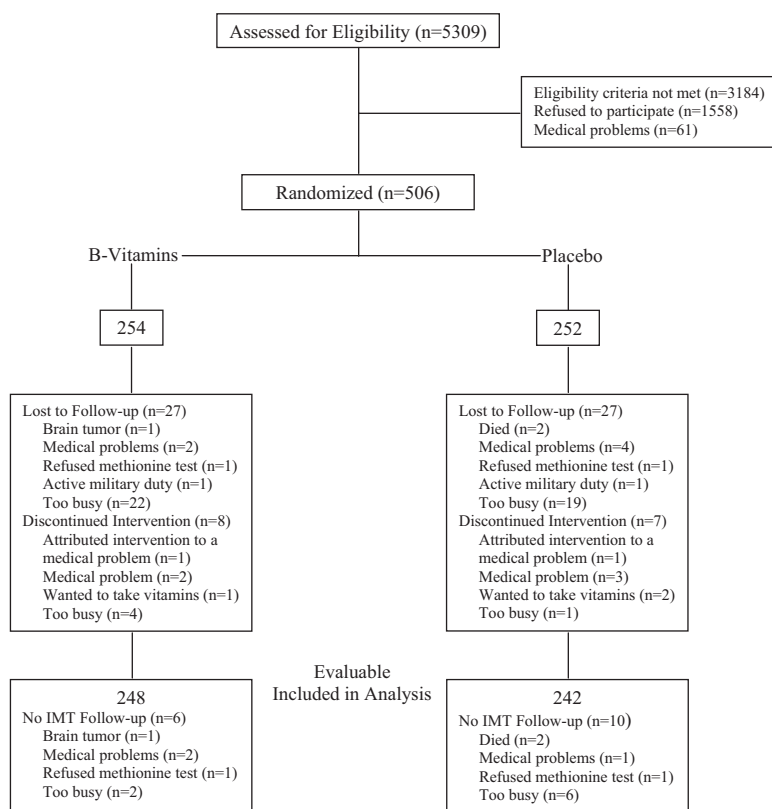


Figure. Trial profile.

cell (WBC) count below the laboratory normal limit (4000 cell/ $\mu$ L) was also compared between treatment groups.

Statistical analyses used SAS 9.0 software (SAS, Inc, Cary, NC); statistical testing was conducted at the 0.05 significance level.

## Results

### Baseline Characteristics

Of the 5309 subjects prescreened by telephone (Figure), 506 were randomized (254 B vitamin, 252 placebo). Of the randomized subjects, 446 (223 B vitamin, 223 placebo) completed the initially planned 2.5-year trial period; 280 (143 B vitamin, 137 placebo) participated in the trial extension. Four hundred ninety subjects (248 B vitamin, 242 placebo) contributed to the primary end point analysis; 443 subjects (224 B vitamin, 219 placebo) had baseline and end-of-trial calcium measures.

Treatment groups did not significantly differ at baseline for demographic, clinical, and atherosclerosis characteristics (Table 1). The average age was 61.4 years, 61% of subjects were male, and 35% were from an ethnic minority.

### Carotid Artery Intima Media Thickness Progression Rates

The 490 subjects with evaluable CIMT data had a mean (range) of 3.14 (0.48 to 4.56) years of follow-up in the B vitamin group and 3.07 (0.46 to 5.0) years of follow-up in the placebo group ( $P=0.63$ ). Participants contributed an average of 8.2 (range, 3 to 11) CIMT measures in the B vitamin group and 8.2 (range, 3 to 11) measures in the placebo group ( $P=0.69$ ). A statistically significant difference between the overall CIMT progression rate in the B vitamin-treated group and the placebo-treated group was not found (Table 2).

However, in a post hoc analysis, among subjects with initial fasting tHcy at or above the median ( $\geq 9.1$   $\mu$ mol/L), the B vitamin-treated group had statistically significant lower average rates of CIMT progression than the placebo-treated group ( $P=0.02$ ); among subjects with a baseline fasting tHcy  $<9.1$   $\mu$ mol/L, there was no significant treatment effect (probability value for treatment interaction=0.02). Subjects with baseline tHcy  $\geq 9.1$   $\mu$ mol/L were older, had lower baseline vitamin B<sub>12</sub> levels, and higher systolic blood pressure than subjects with tHcy  $<9.1$   $\mu$ mol/L (all  $P<0.05$ ).

In mixed-effects models in the entire sample of 490 subjects, baseline and on-trial levels of tHcy, folic acid, vitamin B<sub>12</sub>, and vitamin B<sub>6</sub> were not significantly associated with CIMT progression rate. Both baseline ( $P=0.03$ ) and on-trial PML ( $P=0.01$ ; 2-hour postmethionine tHcy minus fasting tHcy) were positively associated with CIMT progression. On-trial PML was also significantly positively associated with CIMT progression within each treatment group.

### Calcium in the Coronary Artery and Abdominal Aorta Progression

Treatment groups did not differ on baseline CAC and abdominal aortic calcium (Table 3). Changes in calcium measures did not differ between treatment groups overall or by baseline median fasting tHcy. Among subjects who showed no CAC on baseline scan, incidence of new CAC at follow-up was 19% in B vitamin-treated and 17% in placebo-treated subjects ( $P=0.61$ ).

### Homocysteine and B Vitamins

Baseline fasting tHcy, PML, and B vitamin levels did not significantly differ between treatment groups (Table 4).

**Table 1. Baseline Demographic, Clinical, and Atherosclerosis Characteristics**

| Variable                             | B Vitamins<br>(n=254) | Placebo<br>(n=252) | P Value* |
|--------------------------------------|-----------------------|--------------------|----------|
| Age, years                           | 61.7 (10.1)†          | 61.1 (9.6)         | 0.48     |
| Gender                               |                       |                    | 1.00     |
| Male                                 | 155 (61%)             | 154 (61%)          |          |
| Female                               | 99 (39%)              | 98 (39%)           |          |
| Race                                 |                       |                    | 0.76     |
| White                                | 170 (67%)             | 159 (63%)          |          |
| Black                                | 36 (14%)              | 38 (15%)           |          |
| Hispanic                             | 24 (9%)               | 30 (12%)           |          |
| Asian                                | 21 (8%)               | 24 (10%)           |          |
| American Indian                      | 2 (1%)                | 1 (<1%)            |          |
| Education                            |                       |                    | 0.16     |
| High school graduate or less         | 12 (5%)               | 20 (8%)            |          |
| Some college                         | 74 (30%)              | 73 (29%)           |          |
| Trade or business school             | 13 (5%)               | 10 (4%)            |          |
| Bachelor degree                      | 56 (22%)              | 73 (29%)           |          |
| Postgraduate or professional degree  | 94 (38%)              | 76 (30%)           |          |
| Smoking history                      |                       |                    | 0.52     |
| Current                              | 7 (3%)                | 10 (4%)            |          |
| Former                               | 81 (32%)              | 89 (36%)           |          |
| Never smoked                         | 162 (65%)             | 151 (60%)          |          |
| Blood pressure, mm Hg                |                       |                    |          |
| Systolic                             | 128 (14)              | 129 (15)           | 0.45     |
| Diastolic                            | 80 (8)                | 80 (8)             | 0.97     |
| Body mass index, kg/m <sup>2</sup>   | 27.9 (5.0)            | 28.3 (4.8)         | 0.46     |
| Lipids, mg/dL                        |                       |                    |          |
| Total cholesterol                    | 217 (34)              | 217 (33)           | 0.84     |
| Low-density lipoprotein cholesterol  | 133 (31)              | 134 (30)           | 0.81     |
| High-density lipoprotein cholesterol | 58 (16)               | 56 (14)            | 0.12     |
| Triglycerides                        | 128 (60)              | 132 (63)           | 0.44     |
| Apolipoproteins, mg/dL               |                       |                    |          |
| A-1                                  | 149 (24)              | 148 (21)           | 0.51     |
| B                                    | 105 (18)              | 105 (18)           | 0.81     |
| Creatinine clearance, mL/min‡        | 77.5 (22.8)           | 78.2 (24.5)        | 0.73     |
| CIMT, mm                             | 0.75 (0.13)           | 0.76 (0.16)        | 0.42     |
| CAC§                                 | 0 (0–52)              | 0 (0–90)           | 0.23     |
| AC§                                  | 202 (3–1539)          | 253 (0–1561)       | 0.52     |
| Lipid-lowering medications           | 39 (15.4%)            | 36 (14.3%)         | 0.74     |
| Blood pressure medications           | 69 (27.2%)            | 74 (29.4%)         | 0.58     |

\*Group comparisons used the independent Student *t* tests for continuous variables or  $\chi^2$  test for categorical variables. Calcium measures are compared by Wilcoxon rank sum test.

†Mean (SD) or n (%).

(140–age)(weight, kg)

‡Creatinine clearance =  $\frac{(140 - \text{age})(\text{weight, kg})}{72(\text{serum creatinine, mg/dL})(\times 0.85, \text{ if female})}$

§AC indicates abdominal aortic calcium. Calcium measures are median (interquartile range).

**Table 2. CIMT Progression by Treatment Group**

|                                                               | B Vitamins<br>(n=248) | Placebo<br>(n=242) | P Value* |
|---------------------------------------------------------------|-----------------------|--------------------|----------|
| All subjects, adjusted for baseline CIMT strata (<, ≥0.75 mm) | 0.0022 (0.0005)†      | 0.0029 (0.0007)    | 0.31     |
| Baseline fasting plasma total homocysteine                    |                       |                    |          |
| <9.1 $\mu\text{mol/L}$ (n=126/119)‡                           | 0.0028 (0.0006)       | 0.0021 (0.0007)    | 0.41     |
| ≥9.1 $\mu\text{mol/L}$ (n=122/123)‡                           | 0.0016 (0.0007)       | 0.0038 (0.0007)    | 0.02     |

\*P values from mixed-effects models.

†Mean (SE) CIMT progression rate (mm/year).

‡Sample sizes in subgroup analysis (B vitamins/placebo).

Average fasting tHcy and PML levels significantly decreased in B vitamin-treated subjects and increased significantly in placebo-treated subjects ( $P<0.0001$ ). Within both B vitamin-treated and placebo-treated subjects, average plasma B vitamin levels significantly increased from baseline ( $P<0.004$ ). On-trial levels and changes from baseline showed highly significant treatment group differences (all  $P<0.0001$ ) with the B vitamin-treated group demonstrating reduced fasting tHcy and PML levels and increased B vitamin levels relative to the placebo-treated group. B vitamin supplementation did not affect lipid or apolipoprotein levels (data not shown).

### Compliance

Mean (SD) pill compliance was 90.4% (15.0%) among B vitamin-treated subjects and 90.4% (16.0%) among placebo-treated subjects ( $P=0.97$ ).

### Tablet Stability

Folic acid and vitamin B<sub>6</sub> remained stable at 100% of their initial pill content over 3 years. Vitamin B<sub>12</sub> remained stable at 100% of its initial pill content until Month 9; at Months 12, 18, 24, and 36, vitamin B<sub>12</sub> dropped, respectively, to 86%, 77%, 73%, and 66% of its initial pill content. No single pill lot remained in circulation beyond 24 months.

**Table 3. Coronary Artery and Aortic Calcium by Treatment Group**

|                                      | B Vitamins (n=224) | Placebo (n=219) | P Value* |
|--------------------------------------|--------------------|-----------------|----------|
| CAC                                  |                    |                 |          |
| Baseline                             | 0 (0–52)†          | 0 (0–90)        | 0.23     |
| 2.5 years                            | 1.2 (0–115)        | 3.9 (0–166)     | 0.31     |
| Change                               | 0 (0–43)           | 0 (0–52)        | 0.82     |
| Aortic calcium                       |                    |                 |          |
| Baseline                             | 202 (3–1539)       | 253 (0–1561)    | 0.52     |
| 2.5 years                            | 480 (55–2284)      | 486 (12–2684)   | 0.44     |
| Change                               | 224 (20–669)       | 175 (3–737)     | 0.75     |
| Incidence of coronary artery calcium | 24/126 (19%)       | 19/115 (17%)    | 0.61‡    |

\*P values from Wilcoxon rank sum test.

†Median (interquartile range).

‡P value from Fisher exact test.



**Table 4. Plasma Homocysteine and B Vitamin Levels**

|                                                                                   | B Vitamins<br>(n=248) | Placebo<br>(n=242) | P Value Between<br>Treatment Groups |
|-----------------------------------------------------------------------------------|-----------------------|--------------------|-------------------------------------|
| Fasting total homocysteine,<br>$\mu\text{mol/L}$                                  |                       |                    |                                     |
| Baseline                                                                          | 9.5 (2.7)*            | 9.8 (4.4)          | 0.40                                |
| Average on-trial                                                                  | 8.8 (1.8)             | 11.2 (3.8)         | <0.001                              |
| Change                                                                            | -0.7 (2.7)            | 1.4 (3.3)          | <0.001                              |
| P value within group                                                              | <0.001                | <0.001             |                                     |
| Postmethionine loading<br>homocysteine-fasting<br>homocysteine, $\mu\text{mol/L}$ |                       |                    |                                     |
| Baseline                                                                          | 14.7 (6.7)            | 14.8 (6.1)         | 0.88                                |
| Average on-trial                                                                  | 12.8 (4.4)            | 15.7 (6.5)         | <0.001                              |
| Change                                                                            | -2.0 (4.7)            | 0.9 (3.9)          | <0.001                              |
| P value within group                                                              | <0.001                | <0.001             |                                     |
| B vitamins                                                                        |                       |                    |                                     |
| Folic acid, ng/mL                                                                 |                       |                    |                                     |
| Baseline                                                                          | 9.7 (5.7)             | 9.2 (5.0)          | 0.26                                |
| Average on-trial                                                                  | 75.4 (72.7)           | 10.3 (6.3)         | <0.001                              |
| Change                                                                            | 65.6 (72.2)           | 1.2 (6.2)          | <0.001                              |
| P value within group                                                              | <0.001                | 0.004              |                                     |
| $B_{12}$ , pg/mL                                                                  |                       |                    |                                     |
| Baseline                                                                          | 400 (199)             | 394 (143)          | 0.68                                |
| Average on-trial                                                                  | 748 (314)             | 432 (159)          | <0.001                              |
| Change                                                                            | 347 (213)             | 38 (95)            | <0.001                              |
| P value within group                                                              | <0.001                | <0.001             |                                     |
| $B_6$ , pmol/mL                                                                   |                       |                    |                                     |
| Baseline                                                                          | 65 (33)               | 73 (62)            | 0.06                                |
| Average on-trial                                                                  | 350 (131)             | 83 (57)            | <0.001                              |
| Change                                                                            | 285 (120)             | 9 (57)             | <0.001                              |
| P value within group                                                              | <0.001                | 0.01               |                                     |

\*Mean (SD).

P value between treatment groups tests mean group differences by independent *t* tests. P value within treatment group tests significance of change (mean on-trial minus baseline) by paired *t* test.

### Clinical Events

A total of 9 (3.5%) B vitamin-treated and 11 (4.4%) placebo-treated subjects had at least one cardiovascular event during the trial ( $P=0.66$ ). Two placebo-treated subjects died during the trial ( $P=0.25$ ). A total of 31 subjects (16 [6.3%] B vitamin, 15 [6.0%] placebo;  $P=1.00$ ) were diagnosed with cancer during the trial. Of the subjects who had normal WBC at baseline, 35 of 232 (15.1%) B vitamin-treated and 24 of 227 (10.6%) placebo-treated subjects demonstrated at least one instance of low WBC during the trial ( $P=0.16$ ). Infectious illness did not differ between treatment groups (57 [22.2%] B vitamin-treated and 60 [23.8%] placebo-treated subjects reported at least one infectious illness over the trial;  $P=0.72$ ). Treatment groups did not differ in antibiotic use over the trial (28 [11.0%] B vitamin-treated, 41 [16.3%] placebo-treated;  $P=0.09$ ).

### Discussion

Supplementation with combination high-dose folic acid, vitamin  $B_{12}$ , and vitamin  $B_6$  significantly raised the plasma

concentrations of these vitamins and lowered fasting and PML tHcy relative to placebo. However, the high-dose combination of B vitamins did not reduce the progression of CIMT, CAC, or abdominal aorta over a 3.1-year period. Subgroup analysis, however, revealed a statistically significant treatment effect of high-dose combination B vitamins on CIMT progression in individuals with a baseline fasting tHcy  $\geq 9.1 \mu\text{mol/L}$ .

Although the subgroup analysis was done post hoc and the results need confirmation, the findings are consistent with the literature indicating that CVD risk substantially increases when plasma fasting tHcy levels exceed 8 to  $9 \mu\text{mol/L}$ .<sup>6,7</sup> As such, the demonstration of a B vitamin supplementation effect on CIMT in individuals with fasting tHcy  $\geq 9.1 \mu\text{mol/L}$  is highly relevant. Fasting tHcy levels have fallen in the general US population since the US Food and Drug Administration-mandated folate fortification of cereal grains (estimated to increase folate intake 70 to  $120 \mu\text{g/day}$ ) was instituted in January 1998<sup>26</sup> and likely accounts for the lack of baseline folic acid difference between individuals with baseline fasting tHcy levels less than and  $\geq 9.1 \mu\text{mol/L}$ . A folate dosage of 5 mg/day, 50 times the intake of folate from cereal grains, is sufficient to overcome any confounding effects of the US Food and Drug Administration fortification policy because there was a highly significant reduction in fasting tHcy relative to placebo. Subjects with baseline fasting tHcy  $\geq 9.1 \mu\text{mol/L}$  had a significantly lower vitamin  $B_{12}$  level than subjects with a baseline tHcy  $< 9.1 \mu\text{mol/L}$ , possibly contributing to the beneficial effect of B vitamin supplementation on CIMT progression in the former group. Additionally, reduction of on-trial PML with B vitamin supplementation (as a reflection of vitamin  $B_6$  supplementation) also possibly contributed to the reduction in the progression of CIMT relative to the placebo-treated group because on-trial PML was positively associated with CIMT progression. Together, these data suggest that US Food and Drug Administration-mandated folate fortification likely has reduced folate as a substantial risk for the progression of atherosclerosis, whereas vitamin  $B_{12}$  and vitamin  $B_6$  (perhaps through PML levels) remain additional targets for further reducing atherosclerosis progression.

BVAIT indicates that individuals of an average age of 61 years who are at low risk for CVD with a fasting tHcy  $\geq 9.1 \mu\text{mol/L}$  benefit from 3 years of B vitamin supplementation. A similar study indicated that lower B vitamin dosages (daily 2.5 mg folic acid+0.5 mg vitamin  $B_{12}$ +25 mg vitamin  $B_6$ ) over a 1-year treatment period slowed the progression of CIMT relative to placebo.<sup>27</sup> The cohort was selected with a baseline CIMT  $\geq 1 \text{ mm}$  and CIMT progression was significantly dependent on the baseline vitamin  $B_{12}$  concentration.

Randomized, controlled trials of secondary prevention have failed to demonstrate a reduction of CVD with B vitamin supplementation.<sup>10,11</sup> The discordance between BVAIT and observational studies and secondary prevention randomized, controlled trials may be the result of different timing of B vitamin supplementation according to the stage (early versus advanced) of atherosclerosis.<sup>28,29</sup> Consistent with this hypothesis was the fact that B vitamin supplemen-

tation had no effect on CAC or abdominal aortic calcium, markers of late-stage atherosclerosis.

Pill compliance was high and the activity of the folic acid and vitamin B<sub>6</sub> components of the combination pill was stable over the trial. Vitamin B<sub>12</sub> was less stable. The significant rise in plasma B vitamin levels and reduction of fasting tHcy and PML relative to placebo confirm the high level of compliance and pill activity.

Although nonsignificant, the implication of the WBC-lowering effect of high-dose B vitamin supplementation relative to placebo in this trial is unclear. Previous reports indicate a relationship between high WBC levels and CVD and reduction of WBC levels could theoretically contribute to a reduction in atherosclerosis.<sup>30</sup> Safety data failed to indicate any adverse effect of B vitamin supplementation on infections or antibiotic use. An inverse association between plasma B vitamin and WBC levels has been reported from previous observational studies<sup>31</sup>; the mechanism is unknown.

Subjects randomized to BVAIT were at low risk for CVD with baseline plasma B vitamin levels defined as normal based on population distributions. There are no optimally defined plasma B vitamin levels in the context of vascular health. BVAIT indicates that currently defined “normal” plasma B vitamin levels are insufficient for atheroprotection and perhaps a better way to define optimal B vitamin levels for vascular health is reflected through fasting tHcy levels; that is, optimal levels of B vitamins to maintain a fasting tHcy <9.1 μmol/L. Targeting tHcy levels is supported by BVAIT because there was a significant treatment interaction according to baseline tHcy.

The results from BVAIT are limited to individuals without pre-existing CVD. Although B vitamin supplementation reduced the progression of atherosclerosis among subjects with tHcy ≥9.1 μmol/L, BVAIT had insufficient power to statistically compare CVD outcomes.

In conclusion, BVAIT indicates that B vitamin supplementation significantly reduces progression of early-stage subclinical atherosclerosis in well-nourished healthy B vitamin “replete” individuals at low risk for CVD with a fasting tHcy ≥9.1 μmol/L. Further studies to determine whether reducing tHcy levels prevents plaque rupture and clinical events in a population similar to BVAIT are warranted.

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### Disclosures

None.

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